

Nicholas Kelly

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Objective | To advance my education and experience in Computer/Electrical and Software engineering.

Experience

<i>Jun 2023 - Present</i>	CPU Core Architect — Ampere Computing Portland, OR ▪ Micro-architecture development for core L1D, L2, and various other areas within core (e.g. MMU) ▪ Core-wide performance modeling (C++23) and data analysis ▪ Significant contributor for several infrastructure/analysis tools and flows (e.g. CI)
<i>Jun 2016 - Jun 2023</i>	CPU Core Architect — Intel Hillsboro, OR ▪ Micro-architecture development for (inner-core) memory-system and various other areas within core (e.g. front-end) ▪ Core-wide performance modeling/infrastructure (C++17) and data analysis ▪ Primary contributor for several infrastructure/analysis tools used frequently, across teams ▪ Guiding adoption and education of industry-standard SW/CI practices
<i>Jan 2015 - May 2016</i>	Graduate Research Assistant — Prof. Mattan Erez UT Austin Austin, TX ▪ Resiliency characterization through error injection and simulation (C++, Python, Verilog)
<i>May 2015 - Aug 2015</i>	Validation Intern — ARM Austin, TX ▪ Interconnect power and clocking validation/coverage
<i>Jul 2014 - Jan 2015</i>	Post-Silicon Validation Intern — Intel Austin, TX ▪ Validation for emulator debug tools
<i>Jun 2013 - Dec 2013</i>	Electrical Engineering Intern — NACCO Materials Handling Group, Inc. Fairview, OR ▪ Embedded Development (C/C++, ARM, ONFI Flash, SPI/I2C/UART, CANBus, WiFi) ▪ PCB Design, Layout, and Assembly ▪ .NET development (Windows, ASP.NET, C, SQL, CSS, Javascript)
<i>Apr 2012 - Sept 2012</i>	Software Development Intern — Mentor Graphics Wilsonville, OR ▪ Perl, Tcl/tk, and shell scripting; C/C++ development ▪ GUI (Tcl/tk) and CGI (Perl) development
<i>Oct 2012 - Jun 2013</i>	Web Developer, EECS Research Project — Oregon State University Corvallis, OR ▪ Facebook application development ▪ Data visualization and web interface (Javascript, Java servlets)
<i>Jun 2011 - Jun 2013</i>	Student Web Developer — OSU Libraries Corvallis, OR ▪ Development of content (forms, pages, modules) using Drupal/PHP/Javascript ▪ Page design using HTML/CSS ▪ General routine tasks and maintenance work on pages

Qualifications

<i>Computer Arch.</i>	CPU simulation and various forms of data collection using C++17-23 Visualization and analysis in Python/Pandas/Jupyter Scripting with Shell, Python, Ruby, Perl, and Tcl Embedded assembly and/or C development (PIC, AVR, MSP430, ARM) VLSI design with Verilog/SystemVerilog/UVM and various EDA tools
<i>Software</i>	Continuous integration with TeamCity/GitHub/GitLab, for JS/TS, Python, Ruby, and C++ Unit-test frameworks, linting, coverage, and static-analysis within JS/TS, Python, Ruby, and C++ Runtime and memory profiling of C++ programs (VTune, valgrind, jeprof, gprof) Software-engineering practices (e.g. testing, OO, design patterns, etc.) teaching in industry
<i>Electrical</i>	Embedded assembly and/or C development (PIC, AVR, MSP430, ARM) VLSI design with Verilog/SystemVerilog/UVM and various EDA tools Analog circuit simulation (HSPICE, Spectre) Circuit layout for PCBs (Eagle, CircuitMaker) and silicon (Cadence) Coursework in embedded systems, graphics, computer architecture, and analog/digital circuits Knowledge of electrical parts, processes, and troubleshooting
<i>Web Development</i>	Front-end web development, including Javascript/jQuery, CSS/SASS, and HTML5 Back-end web development, including C#, ASP.Net, Perl (CGI), Python, PHP/Drupal, Angular/Typescript/Node.js, and JSP/Servlets Databases, including MSSQL, MySQL, PostgreSQL, SQLite, and MongoDB GUI development, with GTK+, Qt, Tcl/tk, and iOS Game development, with Objective-C (iOS) and Actionscript 2.0/3.0
<i>AI/ML</i>	Usage of common LLMs in performance modeling and tool development Development of MCP servers and skills for usage of specialized tools
<i>Additional</i>	Communication and support skills, across teams Able to learn new material quickly

Education

<i>Jan 2014 - May 2016</i>	University of Texas at Austin — Austin, TX M.S. in Computer Architecture and Embedded Systems (3.92 GPA) - Spring 2016
<i>Sept 2009 - Jun 2013</i>	Oregon State University — Corvallis, OR B.S. in Electrical/Computer Engineering (3.91 GPA) - Spring 2013

Selected Projects

Jan 2014 - June 2016

Computer Architecture and Embedded (UT)

- x86 (subset of ISA) processor in structural-verilog (SystemVerilog, Python, x86)
- Realtime GPU Raytracing
- Lightcuts and Illumination
- An analysis of 3DIC Kogge-stone Adders
- Auto-Multithreading extension for Node.js and V8
- GPU Power virus (genetic algorithm, code generator)
- SDF scheduling genetic algorithm to optimize towards energy usage
- Development of custom RTOS for TI Launchpad (ARM)

Sept 2012 - Jun 2013

VLSI/Analog Design and Simulation Projects (OSU)

- Simulation of power-gating and near-threshold effects on power and delay for XOR gate
- Designed bike POV circuit using SystemVerilog, ModelSim, and Cadence Encounter (Place-and-route)
- Design, simulation (HSPICE/Spectre), and layout (Cadence) of OTAs for different specifications

2009 - Present

Web and Game Development

- "The Wave", activity tracking with Facebook integration (Java servlets)
- "Rundezvous", running/biking/hiking tracking (PHP)
- "Boxarrrific", iOS reaction game (Objective-C)
- "Artisan", iOS drawing/tracking game (Objective-C)
- Various Flash-based games (Actionscript 2.0/3.0)
- Personal websites (C#, ASP.Net; PHP; HTML5, JS, CSS)

Publications

Conferences

- Al-Otoom, D.; **Kelly, N.**; Lindsay, A.; Ahmadi, A.; Kumar, A.; Faubert, A.; Chaffin, B.; Toll, B.; Wallace, E.; Crowthers, L.; Charney, M.; Chin, M.; Spradling, M.; Witscher, S.; Caculo, S.; Fakhri, S.; Kothapalli, V.; Freelove, W.; Madhav, M., "Performance Verification of the AmpereOne® CPU Core" In Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture® (Micro). Athens Greece. November, 2026.
- Chang, C.; Lym, S.; **Kelly, N.**; Sullivan, M. B.; Erez, M., "Evaluating and Accelerating High-Fidelity Error Injection for HPC," In Proceedings of The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). Dallas, TX. November, 2018.
- Meier, R.; **Kelly, N.**; Almog, O.; Chiang, P., "A Piezoelectric Energy-Harvesting Shoe System for Podiatric Sensing" Engineering in Medicine and Biology Society (EMBC), 2014 36th Annual International Conference of the IEEE , pp.622,625,26-30 August 2014.

Workshops

- Chang, C.; Lym, S.; **Kelly, N.**; Sullivan, M. B.; Erez, M., "Hamartia: A Fast and Accurate Error Injection Framework," Workshop on Silicon Errors in Logic-System Effects (SELSE). Boston, MA. April, 2018.

References available on request